

Model Integration in NextGen, NGLAB, and NRDS

DevCon 2026, Salt Lake City, UT
Day 1, Session 1

Josh Cunningham, Jordan Laser, Harsha
Vemula, Quinn Lee Russell, Nia Minor, Gio
Romero, Arpita Patel, Zach Wills, Nels Frazier



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Cooperative Institute for Research
to Operations in Hydrology

Agenda

- **NGIAB Evolution and Fun Facts**
- **NextGen and NGIAB overview**
- **NRDS overview**
- **Hands-on demos**
 - **Running NGIAB**
 - **Integrating a BMI model**



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NGIAB-NRDS Ecosystem Evolution & Fun Facts



Feb 3, 2023

Day One

First commit to
AlabamaWaterInstitute/CloudInfra — the seed
of NGIAB.



4 months

AWI → CIROH-UA

Graduated to CIROH-UA on June 13, 2023.
Arpita's last AWI commit: "Added new repo
location."



6 → 32

Contributors

From AWI to a 25-repo community ecosystem.
Original core still in the top rankings.



141

Releases Shipped

One release every ~9 days across the ecosystem
since launch.



835

Merged PRs

One PR merged every ~1.4 days since February
2023.



35 forks

Community Reach

Teams across the country building their own
workflows on top of NGIAB.

CIROH-UA GitHub Organization

NGIAB-NRDS Repositories

25 repos

CIROH-UA · github.com/CIROH-UA

NGIAB Core	NGIAB Core (cont.)	Community NextGen	NRDS
● NGIAB-CloudInfra ● NGIAB-HPCInfra ● NGIAB-DataPreprocess ● NGIAB-TEEHR ● NGIAB-Client ● NGIAB-Cal ● Community HF Patcher	● ciroh_pyngiab <i>nextgen on 2i2c</i> ● training-NGIAB-101 ● NGIAB-Website ● ngen-cal <i>fork of NOAA-OWP</i> ● NGIAB-Spotpy ● rs_route ● hf_pmtiles	● ngen <i>fork of NOAA-OWP</i> ● t-route <i>fork of NOAA-OWP</i> ● lstm <i>fork of NOAA-OWP</i> ● rust-lstm ● forecast_data_overlay ● nwmurl ● hydrotools <i>fork of NOAA-OWP</i>	● ngen-datastream ● forcingprocessor ● datastreamcli ● nrds-client

NGIAB-NRDS product portfolio website

ngiab.ciroh.org



NGIAB-DataPreprocess
Simplify and accelerate data preparation for NextGen simulations with our intuitive preprocessing tool.



NGIAB-CloudInfra
Deploy NextGen Framework seamlessly in cloud environments with our containerized solution.



NGIAB-HPCInfra
Scale your hydrological modeling with our high-performance computing infrastructure.



NGIAB-TEEHR
Advanced tools for iterative and exploratory analysis of hydrologic model performance.



NGIAB-Visualizer
Interactive visualization platform for exploring hydrological data and model outputs.



NGEN-DataStream
Streamline your hydrologic simulations with our automated workflow orchestration tool.

COMMUNITY IMPACT

Join the growing community of researchers using NextGen In A Box



NGIAB Docker Pulls
awciroh/ciroh-ngen-image

14,985
total pulls

Progress to 15,000 pulls



NGIAB Package Downloads
NGIAB-data-preprocess on Pepy

69,743
all-time downloads

Last 3 Months (Selected Versions)

258

Version filter: 4.6.7, 4.6.6, 4.6.5, 4.6.4, 4.6.3

v4.6.7
97

v4.6.6
45

v4.6.5
45

v4.6.4
36

v4.6.3
35

[View detailed Pepy chart](#)

Workshop setup

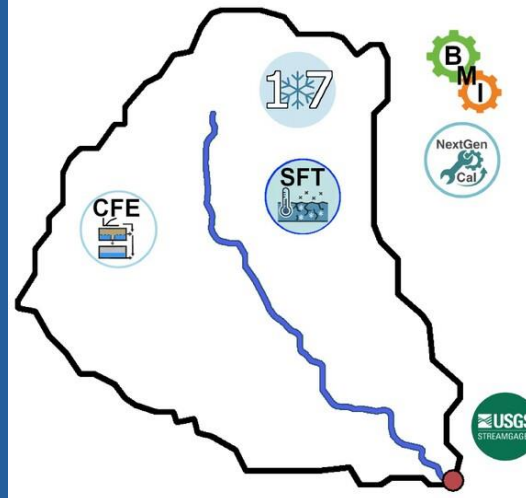
- Please visit <https://github.com/quinnylee/devcon2026-model-int-workshop/blob/main/before-the-workshop.md>



What is NextGen?

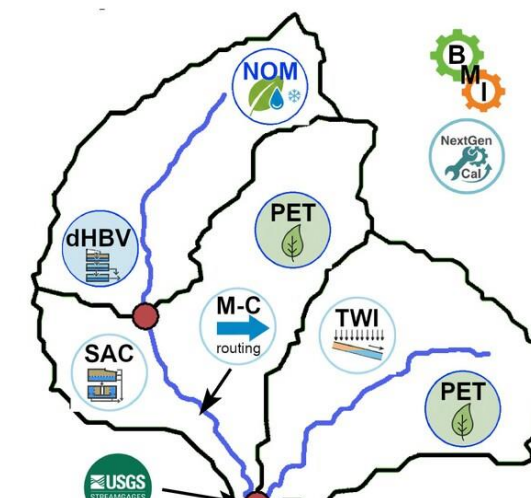
- Next Generation Water Resources Modeling Framework
- Interoperable models
- Uses Basic Model Interface (BMI) to couple models together

a) Single formulation, single catchment, SWE calibration



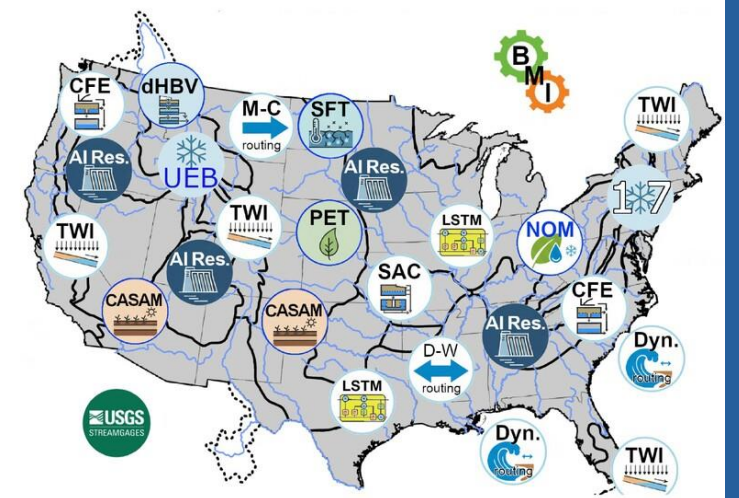
- 17 NOAA Snow-17
- SFT Soil Freeze-Thaw Module
- CFE Conceptual Functional Equivalent to NWM v. 3.1
- BMI NextGen Model Engine Basic Model Interface
- NextGen Evaluation/Calibration Environment
- Hydrofabric Divide
- Hydrofabric Flowpath
- Hydrofabric Nexus

b) Multi-formulation multi-catchment calibration



- PET Potential ET Module Library
- TWI Topographic Wetness Index Model (e.g. Topmodel)
- dHBV Differentiable HBV Model
- SAC Sacramento Soil Moisture Accounting Model
- M-C Muskingum-Cunge hydrologic channel routing
- AI Res. A.I. Informed Reservoir Operations Module
- USGS Discharge Observations

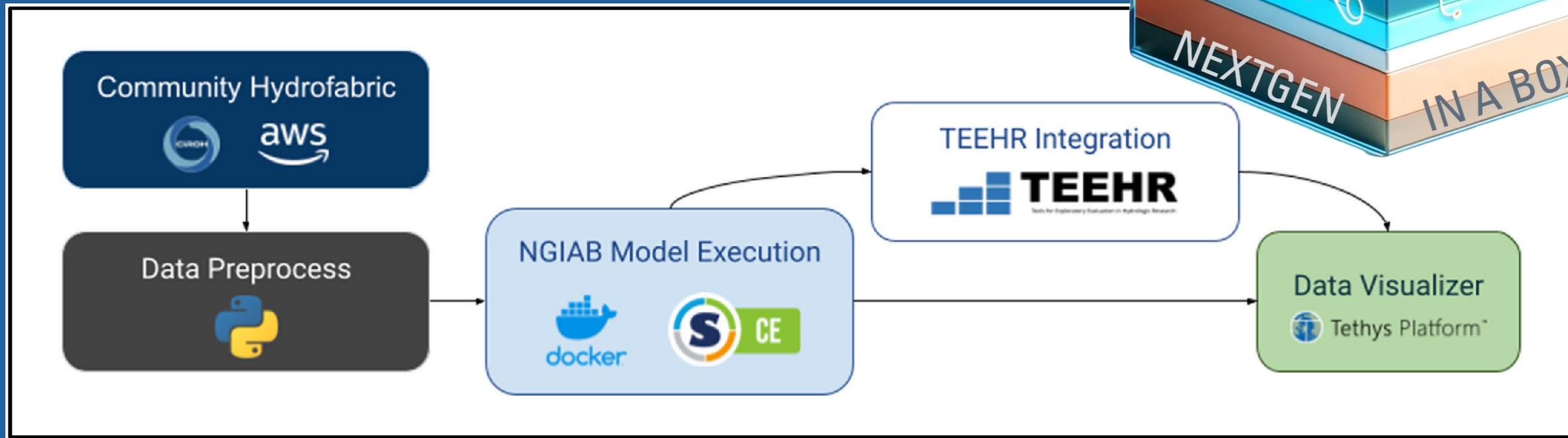
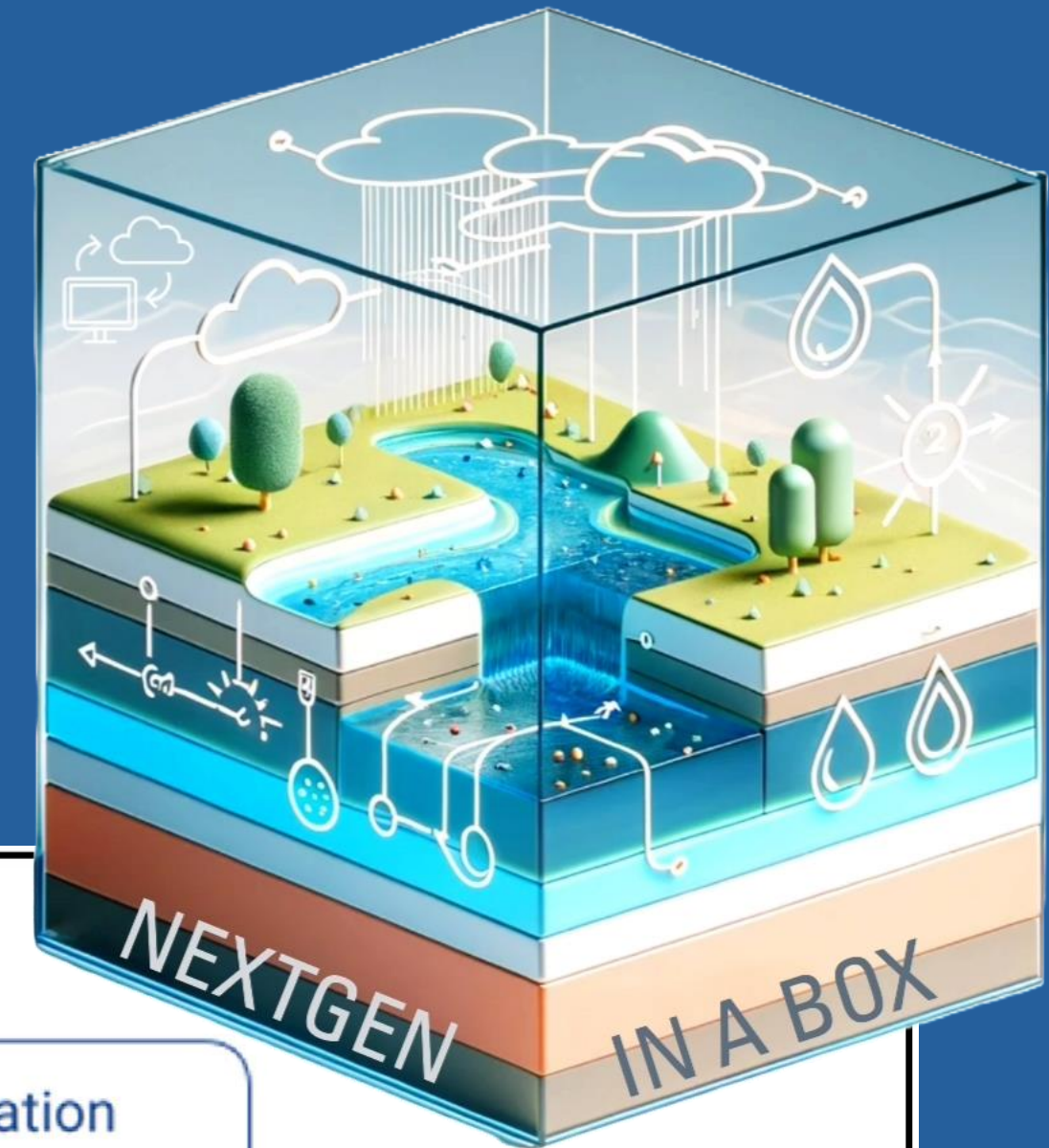
c) Candidate continental scale multi-model mosaic with diverse hydraulic routing methods and AI/ML reservoirs



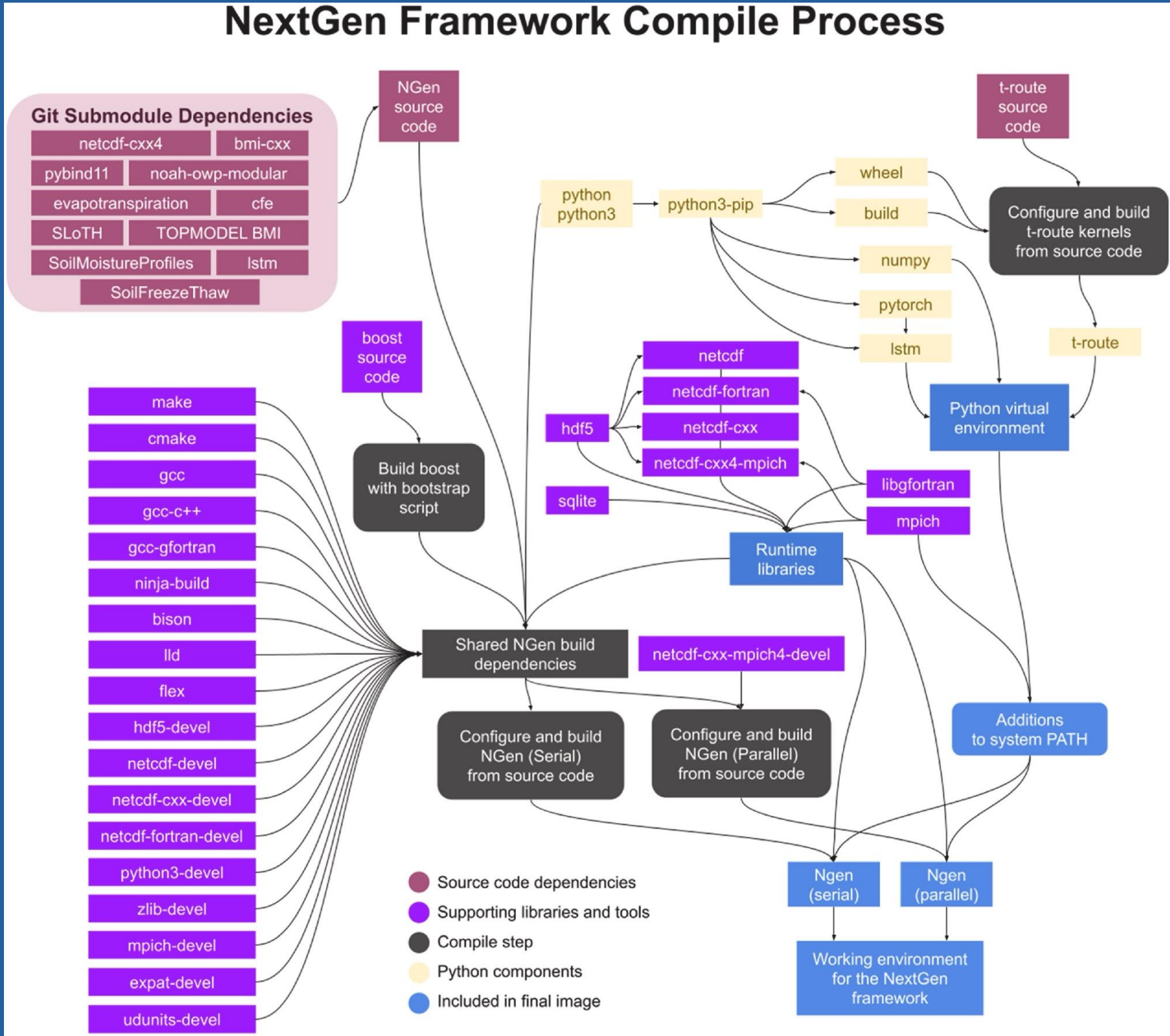
- UEB Utah Energy Balance Snow Melt Module
- NOM NOAA-OWP Modularized Noah-MP Emulator
- LSTM Long Short-Term Memory Deep Learning Module
- CASAM Catchment-Scale Arid/Semiarid Model with Layered Soils
- D-W Diffusive-Wave Hydraulic Routing Module
- Dyn. Dynamic-Wave Hydraulic Routing Module With Coastal Coupling
- Blum et al. (2017) hydroregions
- Major rivers

What is NGIAB?

- Open-source containerized NextGen
- Runs locally, on the cloud, on HPC
- Integrates with TEEHR and Tethys Visualizer



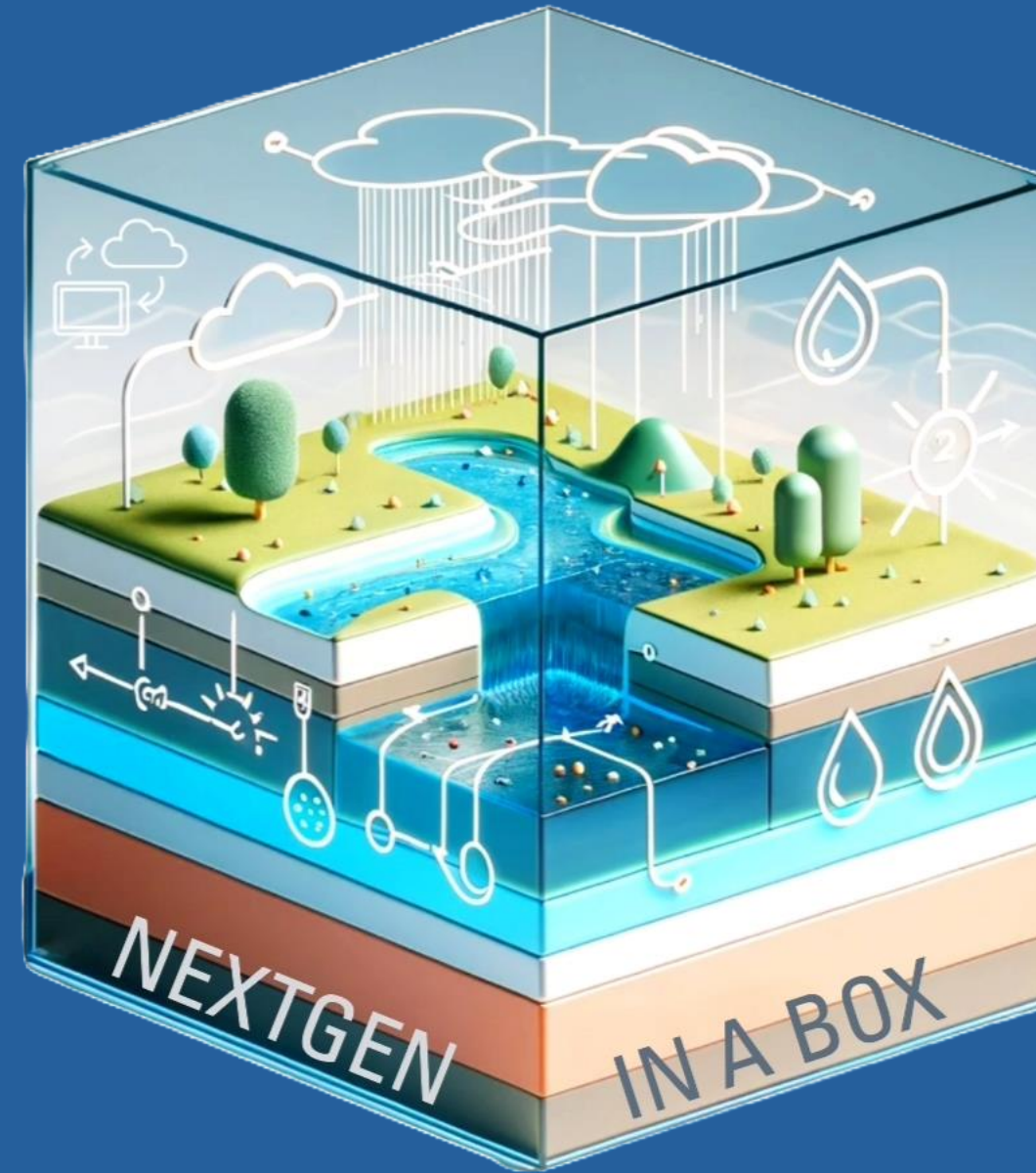
Why NCIAB?



Patel et al. (2025)

Models in NGIAB

- SLOTH
- CFE
- PET
- NOAH-OWP-MODULAR
- TOPMODEL
- LSTM
- Rust LSTM
- dHBV 2.0 (daily and hourly)
- SUMMA
- T-route

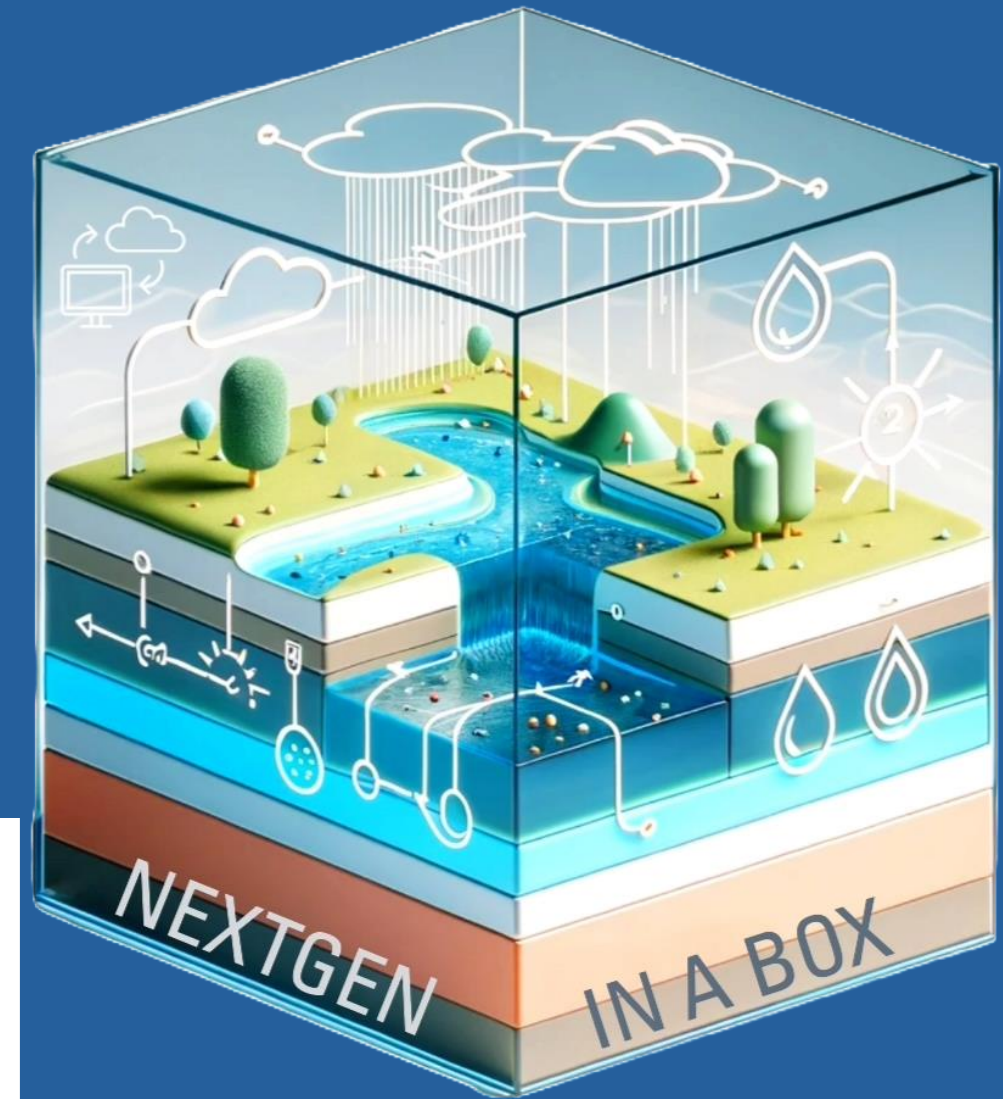


Soon to come...

- SNOW17
- SAC-SMA

Ways to contribute to NGIAB

- Adding a model in ngen/NGIAB
- Add parameters to the hydrofabric
- Add support for more models in the data preprocessor
- Example PRs and technical guidance here (also in workshop repo):



What is the NRDS?

The NextGen Research DataStream (NRDS) is a numerical prediction system for accelerating research to operations by enabling rapid deployment and refinement of new methods in an operational-like testbed.

- 1000+ simulations daily
- Same forecast cadence as NWM
- Several full CONUS datastreams

NRDS Technical
Stack

NextGen Research DataStream [NRDS]

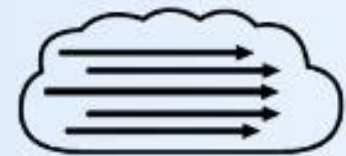
- Daily NextGen simulations across CONUS VPUs.
- Public outputs at datastream.ciroh.org; visualizer at nrds.ciroh.org.



Datastream-1



Datastream-2



Datastream-3

Datastream

- Infrastructure-as-code on AWS
- Scales DataStreamCLI to scheduled, parallel runs with metadata tracking and compute-cost optimization.



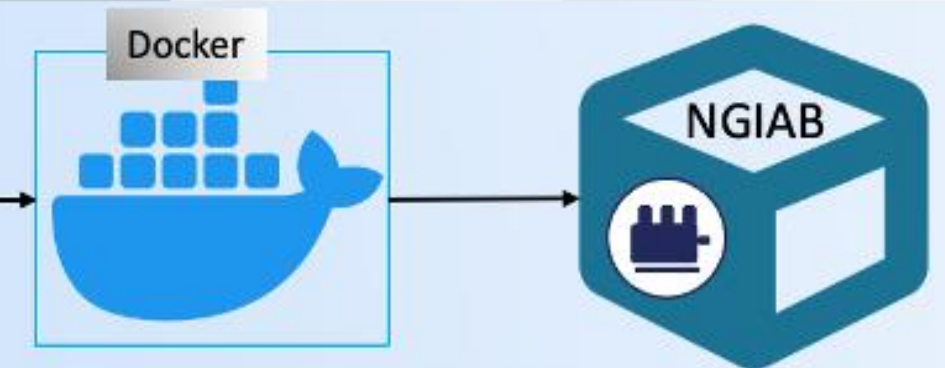
DataStreamCLI

- Stand-alone Command Line Interface (CLI)
- Automates the full pipeline: hydrofabric subset, forcings, NGIAB, and output handling — reproducibly



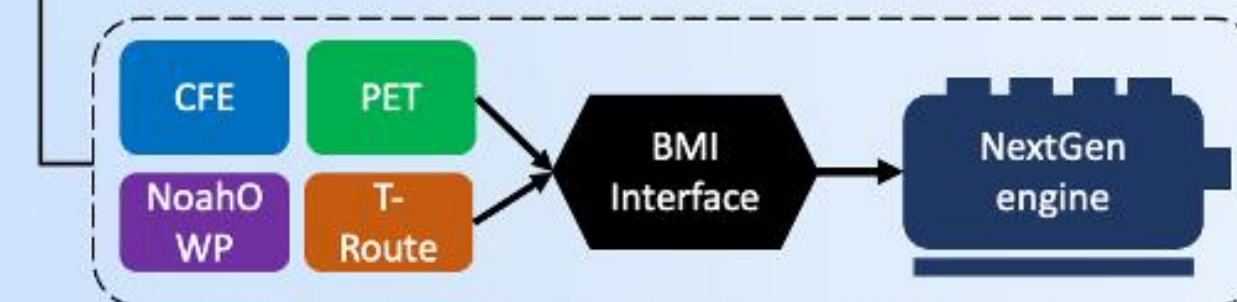
NextGen In A Box [NGIAB]

- Containerized NextGen
- Docker images that package the framework



NextGen Framework

- Modeling engine
- Model/module library



Why the NRDS?

The research goal of the NRDS is to provide open and reproducible infrastructure, tooling, and data to facilitate the refinement of research methods for operational deployment

NRDS Technical
Stack

NextGen Research DataStream [NRDS]

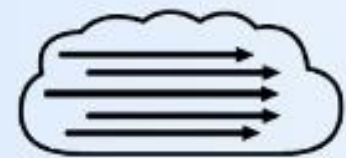
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Datastream-2



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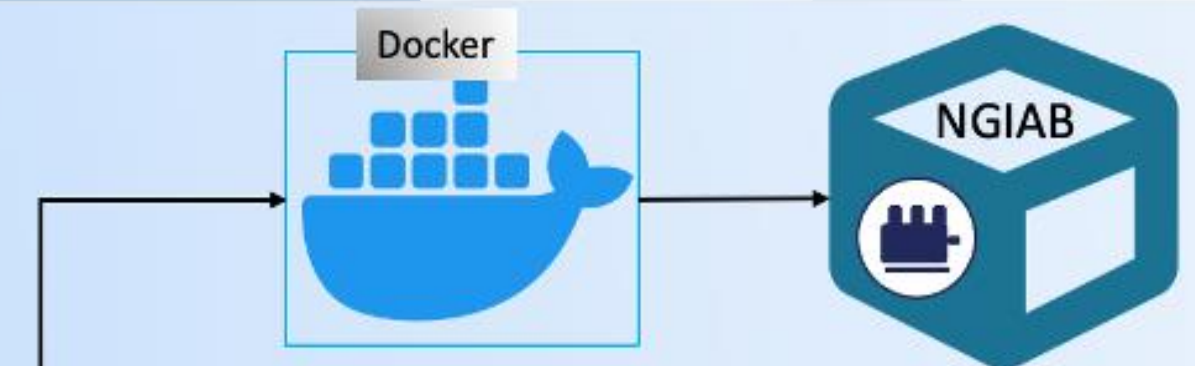
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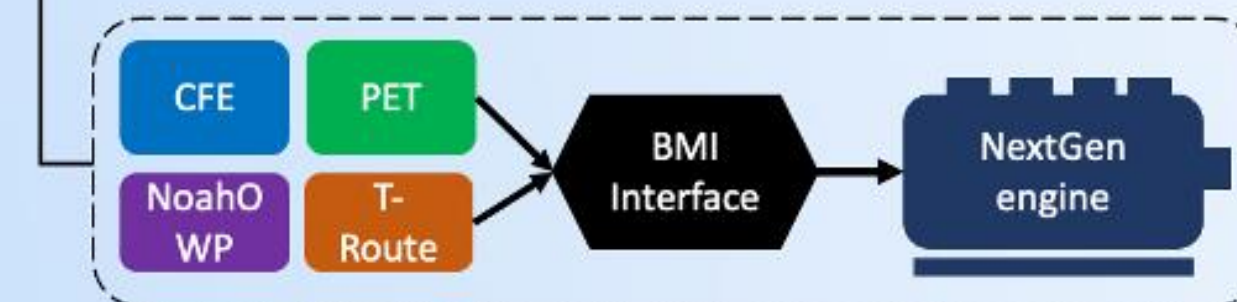
NextGen In A Box [NGIAB]

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NextGen Framework

- Modeling engine
- Model/module library



NRDS: Deployed Datastreams

Name
..
cfe-nom
forcing
Istm_0
qkrig
restart
routing-only

6 existing datastreams

<https://github.com/CIROH-UA/ngen-datastream/tree/main/infra/aws/terraform/services/nrds/datastreams>

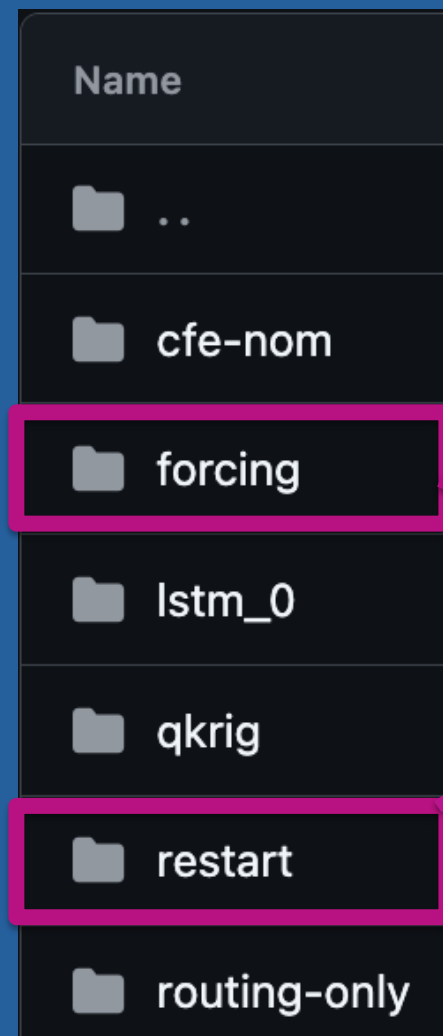
AWS S3 Explorer ciroh-community-ngen-datastream

50 entries per page

Object	Last Modified
backup/	
forcings/	
mappings/	
monitoring/	
outputs/	
parameters/	
realizations/	
resources/	
restarts/	
state/	
status/	
submissions/	
test/	
v2.2_resources/	
index_backup.html	2 years ago
README.html	2 days ago
README.md	2 days ago
s3_info.txt	2 years ago

<https://datastream.ciroh.org/index.html>

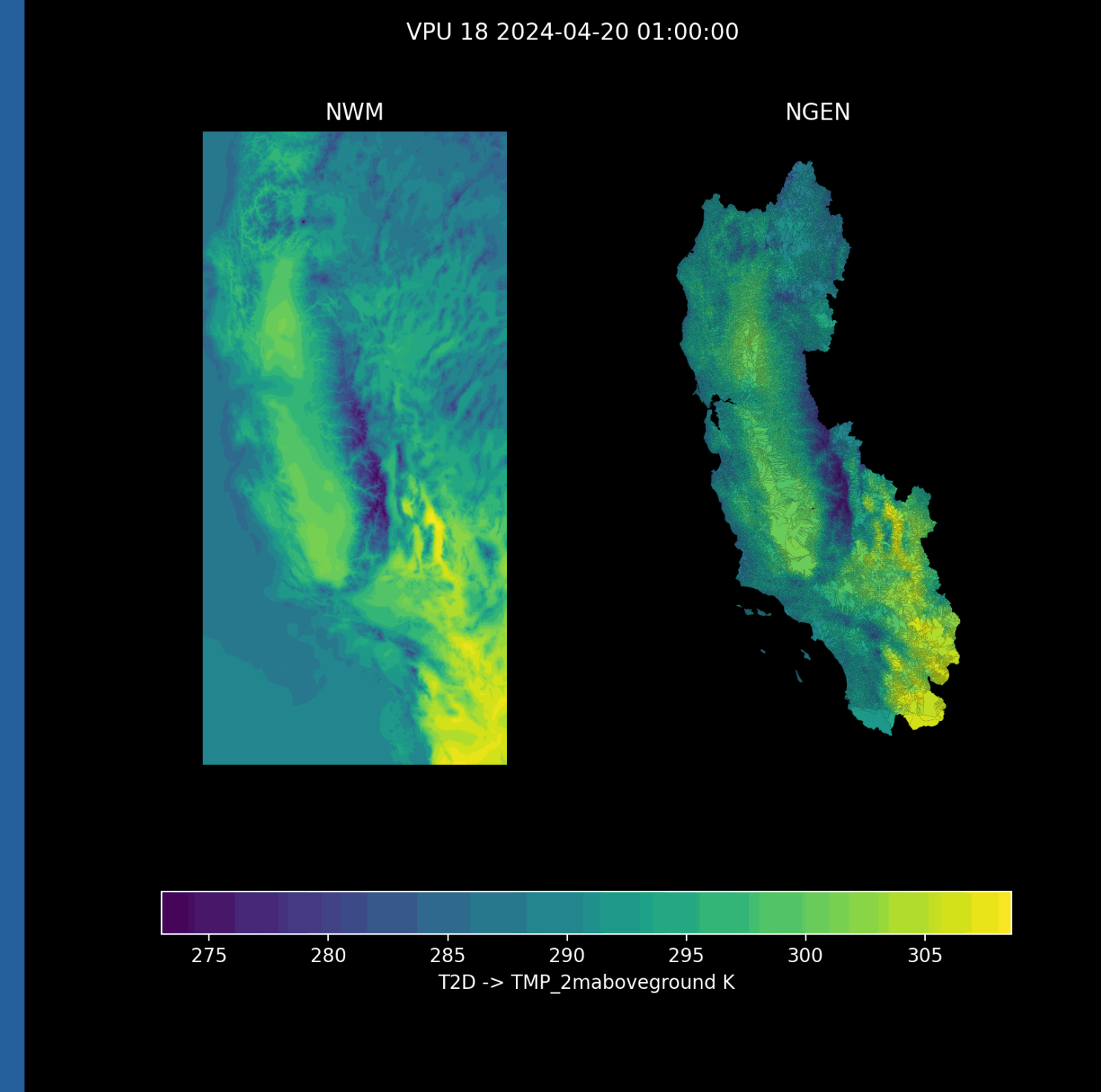
NRDS: Deployed Datastreams



6 existing datastreams

2 preprocess NWM data using the Forcing Processor. The outputs of these datastreams are consumed by other datastreams (composable).

<https://github.com/CIROH-UA/forcingprocessor>



<https://github.com/CIROH-UA/ngen-datastream/tree/main/infra/aws/terraform/services/nrds/datastreams>

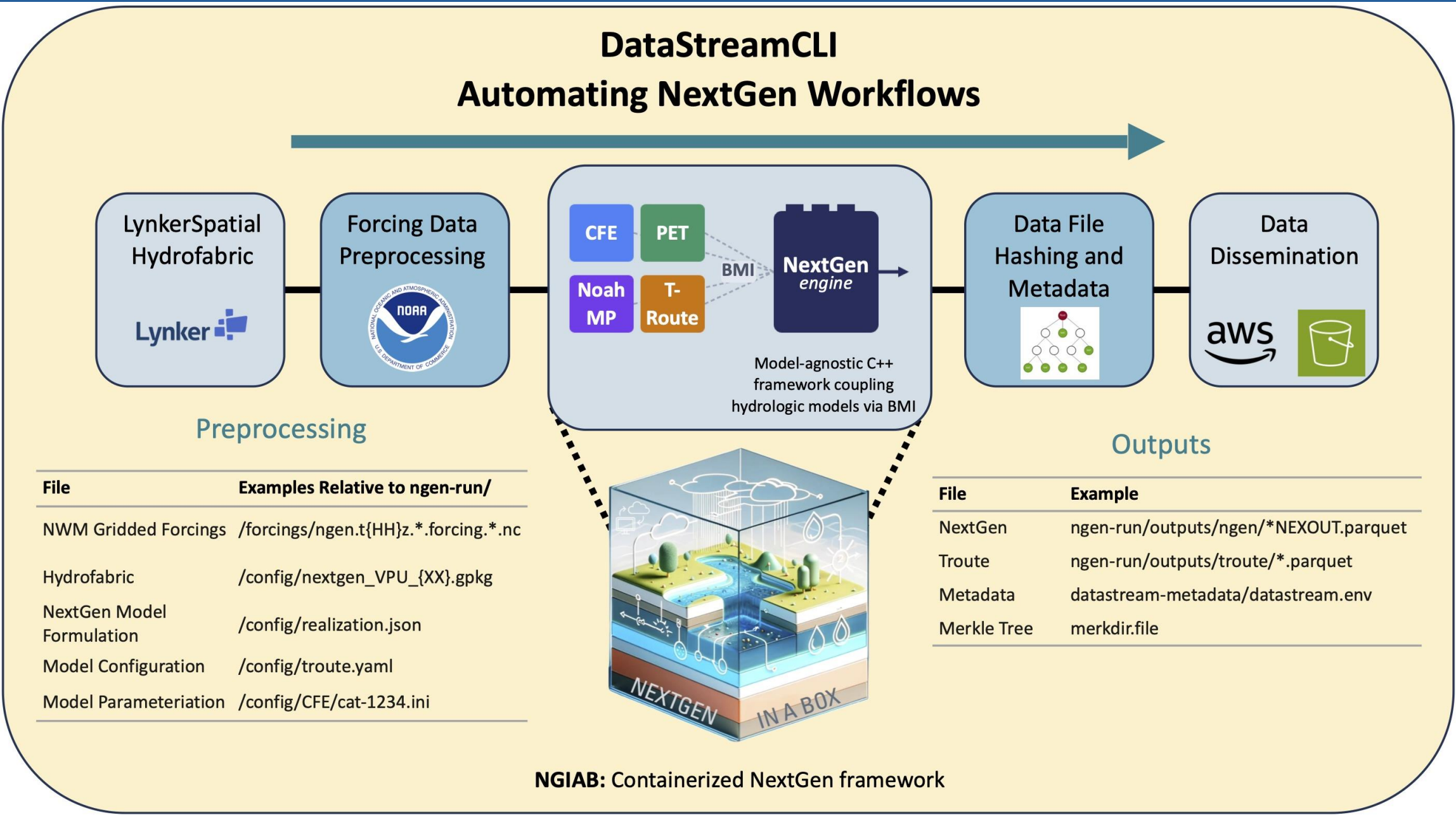
NRDS: Deployed Datastreams



6 existing datastreams

3 are hydrologic forecasts using the NextGen, NGIAB, DataStreamCLI stack. NWM forcings and t-route restarts preprocessed with the forcing and restart datastreams.

<https://github.com/CIROH-UA/ngen-datastream/tree/main/infra/aws/terraform/services/nrds/datastreams>



NRDS: Deployed Datastreams

Name
..
cfe-nom
forcing
lstm_0
qkrig
restart
routing-only

6 existing datastreams

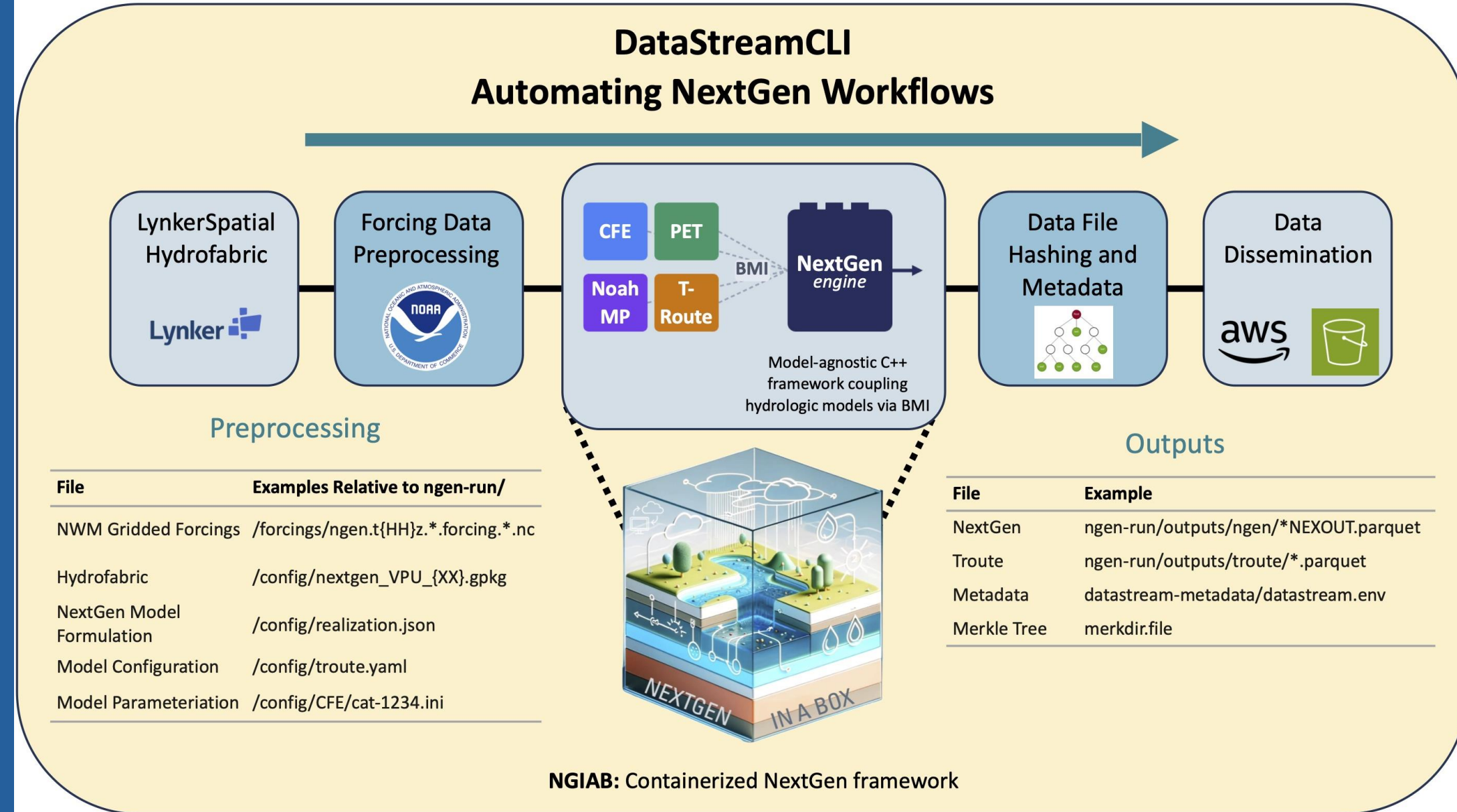
1 is a novel stream flow interpolation method.

** Qkrig community contribution initiated on April 15th, datastream deployed May 20th. 1 month to deployment is revolutionary for R2O

Contributed by Suma Battula and Jonathan Frame

NRDS: How To Reproduce

- DataStreamCLI conducts the hydrologic simulation workflow for all NRDS simulations.
- The open tooling and versioned processing environments enable researchers to reproduce any simulation locally on their laptops.



NRDS Docker Tags

awiciroh/forcingprocessor:1.0.2

awiciroh/datastream:1.0.3

awiciroh/ngen-ciroh-image:1.0.5

<https://github.com/CIROH-UA/forcingprocessor>

<https://github.com/CIROH-UA/datastreamcli>

<https://github.com/CIROH-UA/NGIAB-CloudInfra>

NRDS: How To Reproduce

AWS S3 Explorer

ciroh-community-ngen-datastream / outputs / lstm_0 / v2.2_hydrofabric / ngen.20260522 / short_range / 00 / VPU_01

50 ▾ entries per page

Object
datastream-metadata/
ngen-run/
merkdir.file
ngen-run.tar.gz

Showing 1 to 4 of 4 entries

50 ▾ entries per page

Object
conf_datastream.json
conf_fp.json
conf_nwmurl.json
datastream.env
datastream_ste
docker_hashes.
execution.json
profile.txt
realization_data
realization_use

```
START_DATE="DAILY"
END_DATE="202605220100"
GEOPACKAGE="s3://ciroh-community-ngen-datastream/resources/v2.2_h
NPROCS="7"
NGEN_BMI_CONFS="s3://ciroh-community-ngen-datastream/resources/v2
REALIZATION="https://ciroh-community-ngen-datastream.s3.amazonaws
FORCING_SOURCE="NWM_V3_SHORT_RANGE_00"
NGEN_FORCINGS="s3://ciroh-community-ngen-datastream/forcings/v2.2
FP_TAG=2.0.0
DS_TAG=1.4.0
NGIAB_TAG=v1.7.0
```

Usage: ./scripts/datastream [options]

Either provide a datastream configuration file

-c, --CONF_FILE <Path to datastream configuration file>

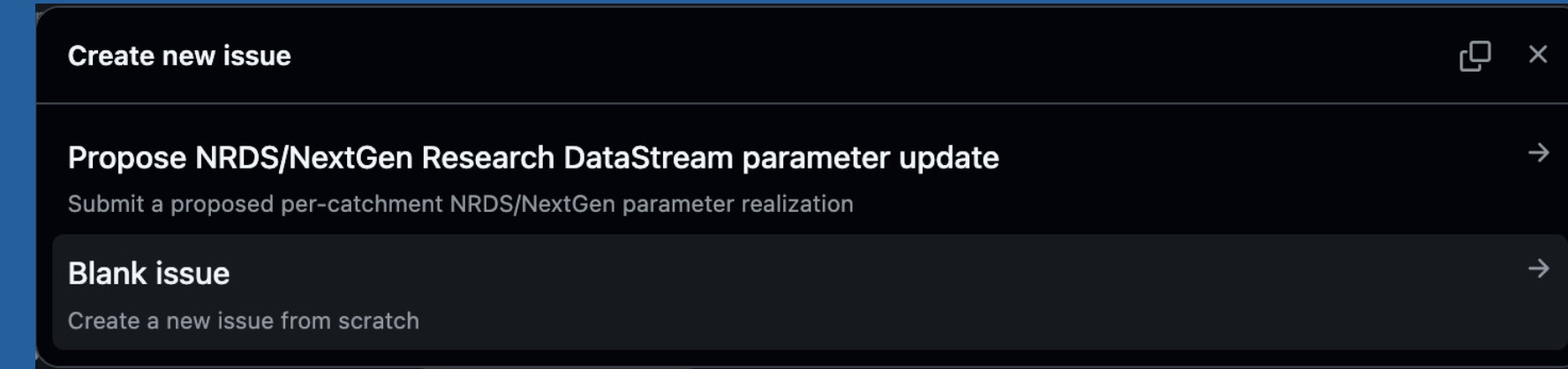
or run with CLI args

```
-s, --START_DATE <YYYYMMDDHHMM or "DAILY">
-e, --END_DATE <YYYYMMDDHHMM>
-C, --FORCING_SOURCE <Forcing source option>
-d, --DATA_DIR <Path to write to>
-R, --REALIZATION <Path to realization file>
-g, --GEOPACKAGE <Path to geopackage file>
-I, --SUBSET_ID <Hydrofabric id to subset>
-i, --SUBSET_ID_TYPE <Hydrofabric id type>
-v, --HYDROFABRIC_VERSION <Hydrofabric version>
-D, --DOMAIN_NAME <Name for spatial domain>
-r, --RESOURCE_DIR <Path to resource directory>
-f, --NWM_FORCINGS_DIR <Path to nwm forcings directory>
-N, --NGEN_BMI_CONFS <Path to ngen BMI config directory>
-F, --NGEN_FORCINGS <Path to ngen forcings directory, tarball, or netcdf>
-S, --S3_BUCKET <s3 bucket to write output to>
-o, --S3_PREFIX <File prefix within s3 bucket>
-n, --NPROCS <Process limit>
-y, --DRYRUN <True to skip calculations>
-E, --EVAL <True to run TEEHR evaluation service>
-L, --LSTM_ENS_MEMBERS <LSTM ensemble members. 012345>
```

NRDS: How To Contribute A DataStream

- Contribute a new datastream
- Contribute an update to an existing datastream
- Considerations

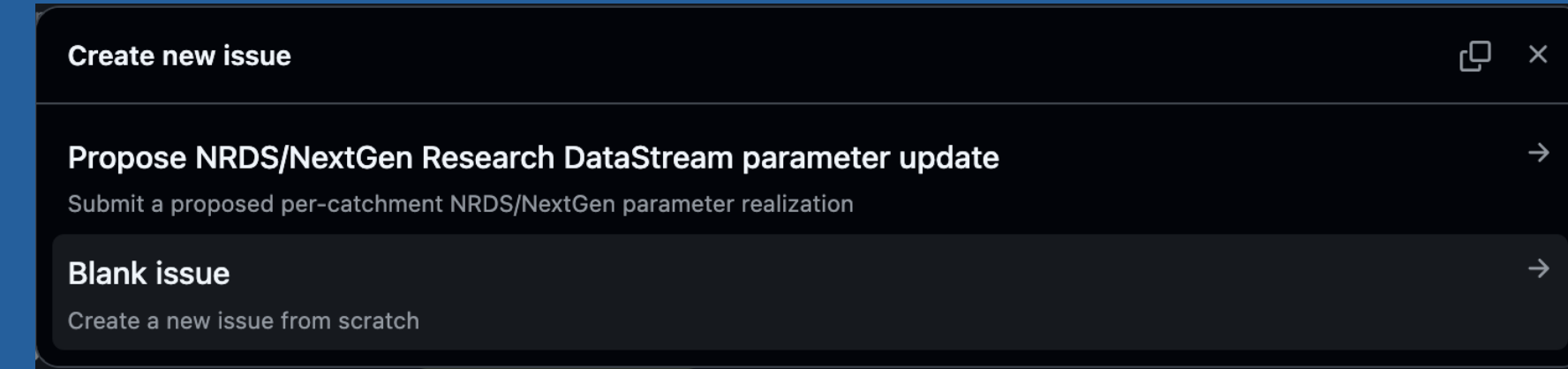
<https://github.com/CIROH-UA/ngen-datastream/issues>



NRDS: How To Contribute A DataStream

- Contribute a new datastream
 - NextGen hydrologic simulation
 - contribute model to NGIAB
 - NextGen non-hydrologic processing
 - Processing gridded NWM files -> Forcing Processor
 - New forcing source (non NWM)
 - For general processing(qkrig), provide code in versioned docker container

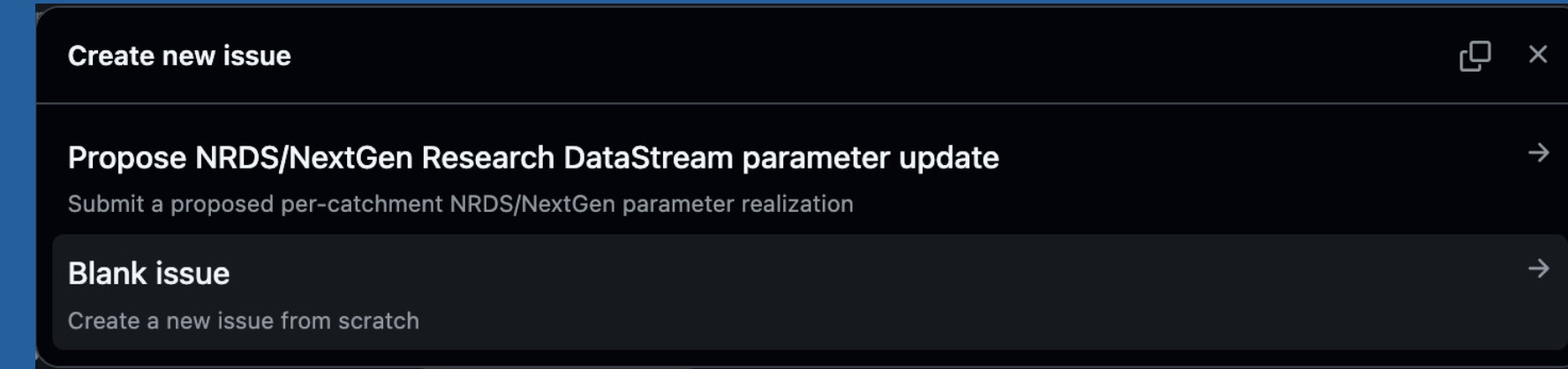
<https://github.com/CIROH-UA/ngen-datastream/issues>



NRDS: How To Contribute A DataStream

- Contribute an update to an existing datastream
 - Updated hydrofabric
 - Calibrated parameter contributions (leaderboard)
 - Updated weights for LSTM
 - Updated model config files (BMI)
 - * Any update that improves the accuracy of outputs or reduces compute cost.

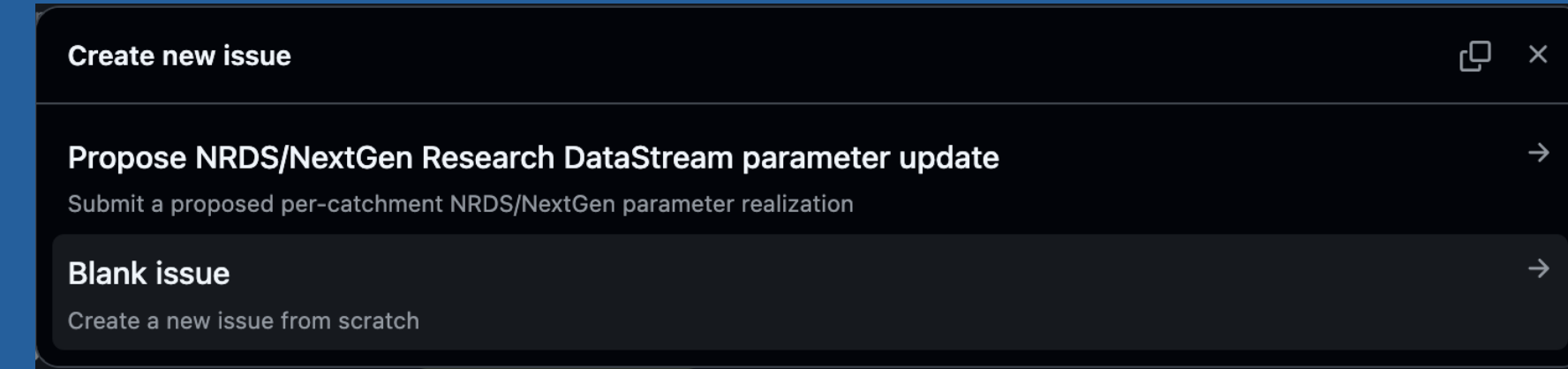
<https://github.com/CIROH-UA/ngen-datastream/issues>



NRDS: How To Contribute A DataStream

- Considerations (think R2O!)
 - Articulate your contribution as best you can in an issue
 - Research value
 - Compute cost
 - Scalability

<https://github.com/CIROH-UA/ngen-datastream/issues>



NRDS: What's Next?

- Continue scaling the NRDS with community contributions (dHBV next!)
- Improve methods leading to more accurate outputs
- Community Contribution Documentation and issue templates
- NRDS Community Governance
- NRDS Leaderboard (clout!)
- DataStreamCLI UI
- Academic article nearing submission

Four ways to run NGIAB

Follow along with the instructions and
code snippets on
<https://github.com/quinnylee/devcon2026-model-int-workshop/blob/main/ngiab.md>



Demo



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Run NGIAB with Data Preprocessor

Demo

> u



Run NGIAB with guide script

Demo

```
> c
```



Run NGIAB with CCNH

Demo

1. Prepare the Python Environment

Import all the required libraries: This section imports the essential Python libraries and packages. In this Jupyter notebook, the `pyngiab.py` script is used to consolidate all the necessary functions for running the NextGen model. More information about the `pyngiab.py` package can be found [here](#).

```
[1]: # ----- Importing Required Libraries -----  
  
from pyngiab import PyNGIAB
```

2. NextGen Run

This section executes the NextGen model using preprocessed data. The user should provide the path to the dataset prepared using the **NextGen Data Preparation** Jupyter notebook.

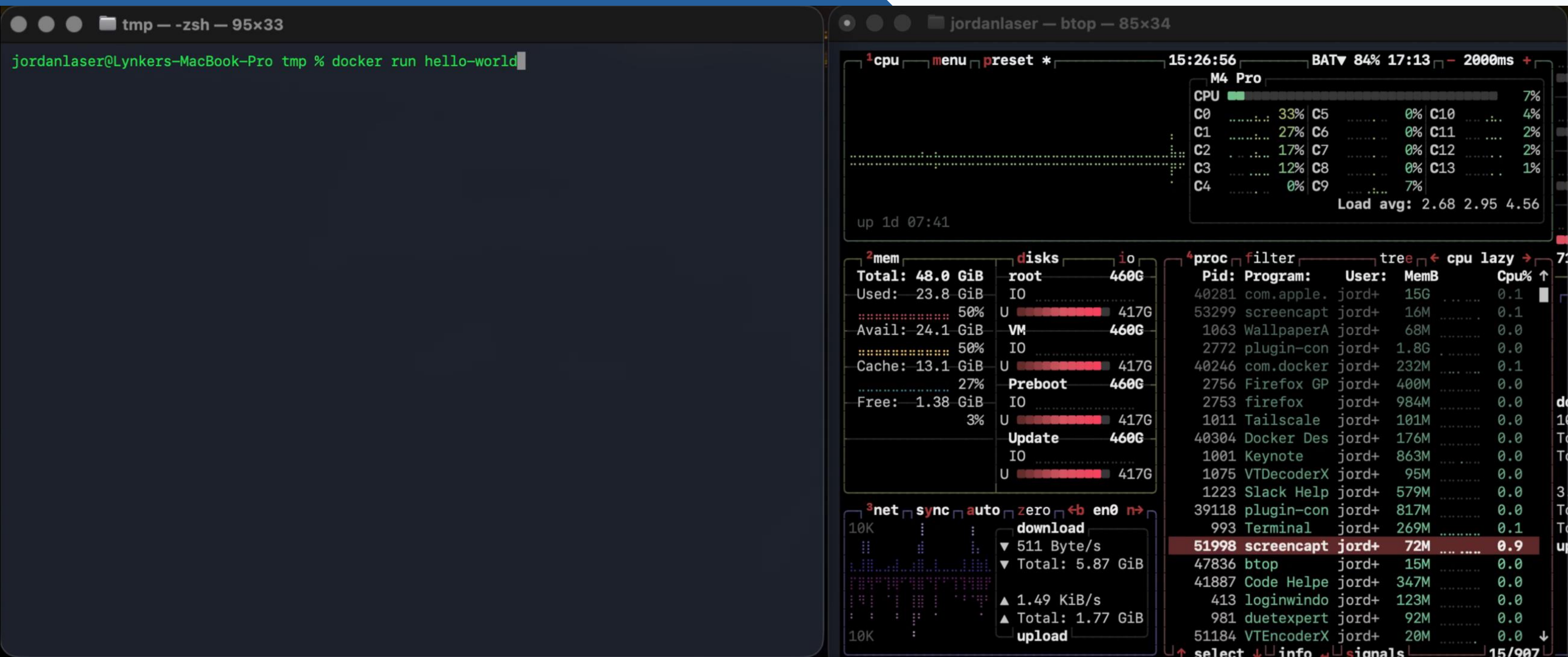
```
[2]: # Specify the HydroFabric subset ID and locate the directory where the NGIAB  
# preprocessing workflow stored the forcing data, configuration files, and realization setup.  
hydrofabric_id = "gage-10109001"  
data_dir = f"/home/jovyan/ngiab_preprocess_output/{hydrofabric_id}"  
  
# Initialize the model for serial execution  
test_ngiab_serial = PyNGIAB(data_dir, serial_execution_mode=False)  
  
# Run the model  
test_ngiab_serial.run()
```

```
WARN: pydantic version: Required(v1), Found(2.7.4).  
WARN: numpy version: Required(1.26.4), Found(1.26.4).  
Required dependencies not found in system path, looking into /ngen/.venv/  
Valid dependencies found in venv: /ngen/.venv/  
*****  
forcings exists. 2 forcings files found.  
config exists. 4 config files found.  
outputs exists. 1 outputs files found.  
Run NextGen Model Framework in Parallel Model ...  
NGen Framework 0.3.0  
NGen Framework 0.3.0  
NGen Framework 0.3.0  
NGen Framework 0.3.0  
NGen Framework 0.3.0  
NGen Framework 0.3.0  
NGen Framework 0.3.0  
NGen Framework 0.3.0  
NGen Framework 0.3.0
```



Reproduce a NRDS Simulation Locally With DataStreamCLI

Demo



Demo

Integrate a model into NGIAB.

See instructions at
[https://github.com/quinnylee/
devcon2026-model-int-
workshop/blob/main/try-it-
intro.md](https://github.com/quinnylee/devcon2026-model-int-workshop/blob/main/try-it-intro.md)



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Thank you!

Contact info:

apatel54@ua.edu

jcunningham8@ua.edu

jlaser@lynker.com

qylee@ua.edu

Workshop resources:

github.com/quinnylee/devcon2026-model-int-workshop



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Join Slack Channel # cirohitsupport



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